01. A software company is developing an online healthcare management system.

For each phase of the Secure Software Development Lifecycle (Requirements, Design, Development, Testing, Deployment, and Maintenance), identify **one security activity** that should be performed and explain why it is important.

01. A developer creates an online shopping application that accepts customer information through web forms.

Identify **five secure coding best practices** that should be followed during development. Explain how each practice helps improve application security.

02. Consider the following user inputs submitted to a registration system.

Input Field	User Input
Name	`John123`
Email	`john.gmail.com`
Age	`-15`
Quantity	`5000`
Username	`<script>alert("XSS")</script>`

For each input:

a) State whether the input should be accepted or rejected.

b) Identify which validation rule is violated (length, type, format, range, or allowed characters).

c) Explain the possible security risk if validation is not performed.

04. An online university portal contains the following users.

- Student
- Lecturer
- Department Head
- System Administrator

For each user role:

a) Identify one authentication mechanism that should be used.

b) List three resources they should be authorized to access.

c) Explain why Role-Based Access Control (RBAC) is suitable for this system.

05. A company is developing an online banking application.

a) Identify **five security-related events** that should be logged.

b) Identify **three sensitive pieces of information** that should never appear in application logs.

c) Explain why secure logging is important during incident investigations.

06. A university develops a Learning Management System (LMS).

Identify the following.

a) Five valuable assets.

b) Five possible threats.

c) Five vulnerabilities.

d) Five possible threat actors.

e) Five possible mitigations.

Present your answer in a table.